

Bobsleigh Contact Geometry

Mass distribution, unilateral contact, and typed observation loss

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Status. This document supersedes the bilingual working draft v1.0. It is migrated to GO Core v0.2 and the Frames–Forces–Dissipation Interface v0.1. Contact and friction are diagnostic smooth-regime models. No sled design, ice constitutive law, safety calculation, or engineering validation is claimed.

Abstract

A center-of-mass trace does not determine the contact state of a sled moving on a shaped surface. We formalize this observation using a regular track parametrization, a finite mass realization, a configuration-space mass form, unilateral gaps, normal reactions, and tangential contact forces. The mass form

$$G_{ij}(q) = \sum_a m_a \partial_i x_a(q) \cdot \partial_j x_a(q)$$

is positive semidefinite and is a metric precisely where the realization map is an immersion. Contact uses the fixed convention

$$g_j \geq 0, \quad N_j \geq 0, \quad N_j g_j = 0, \quad \|T_j\| \leq \mu_j N_j.$$

With T_j defined as the tangential force on the sled and $v_{T,j}$ as its velocity relative to the ice, the nonnegative frictional loss is

$$P_{\text{fric}} = - \sum_j T_j \cdot v_{T,j} \geq 0.$$

Consequently the autonomous stationary-track energy balance is $\dot{E} = P_{\text{act}} - P_{\text{fric}}$ in the smooth active-contact regime. This corrects the sign in v1.0. Cone utilization is defined piecewise at zero normal load, and load entropy has an explicit logarithm base and normalization. Center-of-mass, pose, inertial, contact, load, and thermal channels remain distinct observations of a larger hidden state.

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1 Scope and protocol

Let λ specify a track chart, world and sled frames, clock, contact convention, sampling rates, sensor calibration, estimator, and uncertainty model. The observation chain is

$$X_t \xrightarrow{\mathcal{R}} \tilde{X}_t \xrightarrow{\mathcal{O}_\lambda} Z_t \xrightarrow{\mathcal{K}_\lambda} \text{Law}(Y_t | Z_t) \xrightarrow{\hat{\theta}_\lambda} \hat{\theta}_t.$$

A kinematic centerline supplied as input is not a dynamically predicted trajectory. A proxy load rule is not a solved contact reaction. These statuses are recorded separately.

Boundary 1.1 (Smooth-regime boundary). The equations below apply between impacts while the active contact set is fixed and the state is differentiable. Impact impulses, restitution, Painlevé-type inconsistency, stick-slip switching, and ice deformation require a nonsmooth differential inclusion or complementarity time-stepper and are not proved here.

2 Track as a shaped surface

Let $(e_s, e_u, n_{\text{tr}})$ be an orthonormal track frame and let $\gamma(s)$ be a reference curve. A local trough chart is

$$X_{\text{tr}}(s, u) = \gamma(s) + R_{\text{tr}}(s) \begin{pmatrix} 0 \\ u \\ h(s, u) \end{pmatrix}, \quad R_{\text{tr}} = (e_s, e_u, n_{\text{tr}}) \in \text{SO}(3). \quad (1)$$

Its admissible domain $D \subset \mathbb{R}^2$ must satisfy

$$\partial_s X_{\text{tr}} \times \partial_u X_{\text{tr}} \neq 0 \quad \text{on } D. \quad (2)$$

The surface normal is therefore

$$n_\Sigma = \frac{\partial_s X_{\text{tr}} \times \partial_u X_{\text{tr}}}{\|\partial_s X_{\text{tr}} \times \partial_u X_{\text{tr}}\|}.$$

Centerline curvature, banking, transverse trough curvature, and chart regularity are distinct data. A centerline alone does not determine contact geometry.

3 Finite mass realization

Let a rigid sled be represented by masses $m_a > 0$ at body points r_a^B . Its total mass, center of mass, and inertia about the center of mass are

$$M = \sum_{a=1}^n m_a, \quad r_{\text{cm}}^B = \frac{1}{M} \sum_{a=1}^n m_a r_a^B, \quad (3)$$

$$I_B = \sum_{a=1}^n m_a \left(\|\rho_a\|^2 I_3 - \rho_a \rho_a^\top \right), \quad \rho_a = r_a^B - r_{\text{cm}}^B. \quad (4)$$

For world origin $x^I(t)$ and attitude $A_{I \leftarrow B}(t) \in \text{SO}(3)$,

$$x_a^I = x^I + A_{I \leftarrow B} \rho_a.$$

Let $q \in Q$ be local generalized coordinates after a declared reduction, and let $\Phi(q) = (x_1(q), \dots, x_n(q))$. Define

$$G_{ij}(q) = \sum_{a=1}^n m_a \partial_i x_a(q) \cdot \partial_j x_a(q). \quad (5)$$

Proposition 3.1 (When the mass form is a metric). *$G(q)$ is positive semidefinite. It is positive definite if and only if $d\Phi_q$ is injective.*

Proof. For $\xi \in T_q Q$,

$$\xi^\top G(q) \xi = \sum_a m_a \|d\Phi_q(\xi)_a\|^2.$$

□

Thus redundant coordinates produce a degenerate mass form. Calling it a Riemannian metric requires the immersion hypothesis.

4 Unilateral contact and force pullback

For candidate contact j , let $g_j(t, q)$ be the signed gap, with $g_j > 0$ meaning separation. In the smooth force regime:

$$g_j \geq 0, \quad N_j \geq 0, \quad N_j g_j = 0. \quad (6)$$

Let n_j^I be the ice-to-sled unit normal and $P_{T,j} = I - n_j^I (n_j^I)^\top$. The contact force on the sled is

$$F_j^I = N_j n_j^I + T_j^I, \quad T_j^I \cdot n_j^I = 0. \quad (7)$$

If $J_j^I = \partial_q c_j^I$ is the contact-point Jacobian, its generalized force is

$$Q_j = J_j^\top F_j^I \in T_q^* Q.$$

This pullback is required before adding Q_j to other generalized forces.

Let

$$v_{T,j}^I = P_{T,j} (v_{\text{sled},j}^I - v_{\text{ice},j}^I).$$

A Coulomb-type admissibility condition is

$$\|T_j^I\| \leq \mu_j N_j, \quad T_j^I \cdot v_{T,j}^I \leq 0. \quad (8)$$

For sliding, a maximum-dissipation selection is

$$T_j^I = -\mu_j N_j \frac{v_{T,j}^I}{\|v_{T,j}^I\|}.$$

For sticking, $v_{T,j} = 0$ and T_j is set-valued inside the cone.

5 Dynamics and corrected energy balance

In a smooth reduced chart, let

$$L(t, q, \dot{q}) = \frac{1}{2} \dot{q}^\top G(t, q) \dot{q} - V(t, q).$$

The Lagrange-d'Alembert equation is

$$\frac{d}{dt} \partial_{\dot{q}} L - \partial_q L = Q_{\text{act}} + Q_{\text{fric}} + Q_{\text{con}}. \quad (9)$$

With

$$E_L = \partial_{\dot{q}} L \cdot \dot{q} - L,$$

the frame-force interface gives

$$\dot{E}_L = P_{\text{act}} - P_{\text{fric}} + P_{\text{con}} - \partial_t L, \quad (10)$$

where the nonnegative frictional dissipation is

$$\boxed{P_{\text{fric}} = - \sum_j T_j^I \cdot v_{T,j}^I \geq 0.} \quad (11)$$

Corollary 5.1 (Stationary ideal support). *If the track is stationary, normal constraints are ideal, L is autonomous, and no other nonconservative forces act, then between impacts*

$$\dot{E}_L = -P_{\text{fric}} \leq 0.$$

Remark 5.2 (Correction to v1.0). Version v1.0 defined T_j as the tangential force on the sled but wrote $\dot{E} = -P_{\text{fric}}$ with $P_{\text{fric}} = \sum_j T_j \cdot v_{j,\text{rel}}$. Under dissipative friction that sum is nonpositive, so the displayed energy law increased energy. Equation equation (11) defines the loss with the required leading minus sign.

For a moving support, normal reactions can exchange power and P_{con} need not vanish. The same support work must not also be inserted through a reduced time-dependent L .

6 Guarded diagnostics

6.1 Cone utilization

Define

$$u_j = \begin{cases} \|T_j\| / (\mu_j N_j), & \mu_j N_j > 0, \\ 0, & \mu_j N_j = 0 \text{ and } T_j = 0. \end{cases}$$

The cone law excludes $\mu_j N_j = 0, T_j \neq 0$. For a finite candidate set J ,

$$S_{\text{slip}} = 1 - \max_{j \in J} u_j, \quad \max \emptyset := 0.$$

Then $0 \leq S_{\text{slip}} \leq 1$ on admissible states. This is a cone margin, not a probability of slipping.

6.2 Load distribution and entropy

If $N_{\text{tot}} = \sum_{j=1}^K N_j > 0$, set

$$p_j = \frac{N_j}{N_{\text{tot}}}, \quad H_b(N) = - \sum_{j:p_j>0} p_j \log_b p_j, \quad b > 1.$$

For a declared $K \geq 2$, the normalized diagnostic is

$$\overline{H}_b = \frac{H_b(N)}{\log_b K} \in [0, 1].$$

It is independent of the logarithm base. If $N_{\text{tot}} = 0$, the load distribution and entropy are undefined rather than silently set to zero.

6.3 Two-runner imbalance

For $N_L + N_R > 0$,

$$\Delta_N = \frac{|N_R - N_L|}{N_R + N_L} \in [0, 1].$$

The diagnostic is undefined in free flight.

6.4 Geometric demand and banking mismatch

For a centerline with curvature κ and speed v ,

$$a_\kappa = v^2 \kappa$$

has acceleration dimension. If $f_{\text{req}} = m(a - g) \neq 0$, define the unsigned mismatch

$$\Delta_b = \text{atan2}(\|n_{\text{tr}} \times f_{\text{req}}\|, n_{\text{tr}} \cdot f_{\text{req}}) \in [0, \pi].$$

This is a directional diagnostic; it does not solve the contact forces.

7 Observation channels

The hidden record may contain

$$X_t = (\Sigma, \{m_a, r_a^B\}, q, \dot{q}, \mathcal{A}_t, \{c_j, N_j, T_j\}_{j \in \mathcal{A}_t}, \vartheta),$$

where $\mathcal{A}_t$ is the active contact set and ϑ collects constitutive parameters. Useful deterministic channels include

$$\begin{aligned} \mathcal{O}_{\text{cm}}(X_t) &= x_{\text{cm}}^I(t), & \mathcal{O}_{\text{pose}}(X_t) &= A_{I \leftarrow B}(t), \\ \mathcal{O}_{\text{imu}}(X_t) &= (f^B, \Omega^B), & \mathcal{O}_{\text{load}}(X_t) &= (N_j)_{j \in \mathcal{A}_t}, \\ \mathcal{O}_{\text{contact}}(X_t) &= (c_j)_{j \in \mathcal{A}_t}, & \mathcal{O}_{\text{heat}}(X_t) &= P_{\text{fric}}(t). \end{aligned}$$

No equality between these channels is implied.

Proposition 7.1 (Conditional center-of-mass non-identifiability). *Assume the admissible hidden-state class contains two records $X_t^{(1)} \neq X_t^{(2)}$ with the same center-of-mass*

curve but different mass realization or contact record. Then $\mathcal{O}_{\text{cm}}$ is not injective on that class.

Proof. By hypothesis, $\mathcal{O}_{\text{cm}}(X_t^{(1)}) = \mathcal{O}_{\text{cm}}(X_t^{(2)})$ while the arguments differ. $\square$

This proposition is deliberately conditional. Showing that two specific sleds admit the same trajectory under a fixed constitutive model is a separate dynamical existence problem.

8 Reproducibility record

A numerical benchmark is admissible only if it reports:

- (i) the surface chart and its regularity domain;
- (ii) mass points, center of mass, inertia, and coordinate chart;
- (iii) active-set and contact-law solver;
- (iv) ice-motion convention and friction parameters;
- (v) integration method, tolerances, event handling, and convergence study;
- (vi) sensor map, sampling, noise, and estimator;
- (vii) which outputs are solved reactions and which are proxies.

The proxy values in v1.0 are not retained as validated mechanics because no complementarity/contact solver or convergence study was attached.

9 Claim boundary

The strict result is a corrected and typed interface to standard constrained mechanics. It does not claim:

- a predictive ice-runner friction law;
- uniqueness of the contact solution;
- engineering accuracy or safety certification;
- physiological rider limits;
- a new theorem beyond the stated interface propositions.

References

- [1] B. Brogliato, *Nonsmooth Mechanics: Models, Dynamics and Control*, 3rd ed., Springer, 2016. doi:10.1007/978-3-319-28664-8.
- [2] A. M. Bloch, *Nonholonomic Mechanics and Control*, 2nd ed., Springer, 2015. doi:10.1007/978-1-4939-3017-3.
- [3] V. Acary and B. Brogliato, *Numerical Methods for Nonsmooth Dynamical Systems*, Springer, 2008. doi:10.1007/978-3-540-75392-6.